

Shrivatsa Deshmukh

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Summary: Computational neuroscience researcher with a background in robotics and physics, applying Bayesian and statistical modeling to neural data across scales, from single neurons to whole-brain systems, to study neural coding, circuit dynamics, and disease-related dysfunction.

SKILLS

Neuro & Computational: EEG, fMRI, Calcium Imaging, Neural Data Analysis, Simulation-Based Inference (SNPE), Hierarchical Statistical Modeling (GLMMs), MNE-Python

Programming & ML: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-learn, MATLAB, Git, Linux/Bash

EXPERIENCE

Noel Lab, University of Minnesota | Research Associate

November 2025 – Present

- Designed a Zero-Inflated Negative Binomial (ZINB GLMMs) pipeline to characterize ASD-associated neural signatures across three rodent models (Fmr1, Cntnap2, Shank3 KOs) and wildtype controls in International Brain Laboratory (IBL) electrophysiological data spanning 32 brain regions.
- Quantified neural response variability, contrast response functions, and adaptation dynamics across 613 neurons and 375,000+ trials; preliminary ASD vs. wildtype differences align with existing literature, with cross-model analysis ongoing.

Brain Networks Group, Max Planck Institute CBS | Guest Researcher

May 2025 – September 2025

- Implemented amortized Bayesian inference for a Jansen-Rit neural mass model via sequential neural posterior estimation (SNPE), training a Neural Spline Flow on 32,768 biophysically-grounded EEG simulations (MNE fsaverage lead-field, per-trial 1/f noise); $R^2 = 0.79$, 90% CI coverage 0.83–0.89.
- Ran a retrain-based ablation study showing two jointly-trained gating mechanisms (sigmoid, softmax) are negatively rank-correlated with true feature importance (Spearman $\rho \approx -0.55, -0.45$, 10 seeds) despite agreeing closely with each other ($\rho \approx 0.83$); extending the attention-vs-explanation gap from NLP/EEG deep learning to mechanistic SBI.

Alfonso Araque Lab, University of Minnesota | Research Associate

March 2025 – November 2025

- Built a Python pipeline for AQuA2 astrocyte calcium imaging data; automated per-slice fold-change normalization and quantile regression (Q50/Q75/Q90) across 16 spatiotemporal and network features, 4 experimental groups, and 96 recording sessions.
- Demonstrated that psilocybin selectively amplifies astrocyte network synchrony and event size through the 5-HT_{2A}/IP3R2 signaling pathway, with effects concentrated in the upper tail (Q90 suppression $\sim 6.5\times$ FC in antagonist and knockout groups vs near-zero at Q50); within-group analysis additionally revealed a pathway-independent increase in integrated calcium signal and event duration, a finding invisible to standard between-group comparison.

Health Intelligence Lab, University of Minnesota | Research Associate

November 2024 – March 2025

- Decomposed fMRI data from ~ 100 subjects via tensor decomposition to examine amygdala reactivity during an emotion processing task; found unexpectedly weak amygdala-BDI (Beck Depression Inventory) correlations despite the amygdala's established role in emotion processing.
- Explored dynamic mode decomposition (DMD) and 6 variants via PyDMD on resting-state EEG; identified Optimized DMD as most noise-robust and characterized mode incoherence across time windows as the fundamental limit of long-horizon decomposition.

Indian Institute of Technology Kanpur | Machine Learning Researcher

January 2023 – August 2023

- Designed a pipeline decoding motor intent from EMG signals for prosthetic and exoskeleton hand control, integrating biological signal processing with computer vision outputs. Multilayer Perceptron (MLP) achieved 11.57% accuracy improvement over prior methods.
- Publication:** Deshmukh Shrivatsa, Vitthal Khatik, and Anupam Saxena. "Robust Fusion Model for Handling EMG and Computer Vision Data in Prosthetic Hand Control." *IEEE Sensors Letters* (2023). doi:10.1109/LSENS.2023.3301837

PROJECTS

Neural Phoneme Decoder for Speech BCI

January 2025 – May 2025

- Reproduced gated recurrent unit (GRU) phoneme decoder from Willett et al. (2023, Nature); isolated the decoder stage from the full Kaldi pipeline and reduced compute requirements via hyperparameter optimization to enable single-GPU training, achieving 78.77% phoneme accuracy within 1.5% of the published benchmark.

Spatiotemporal EEG Decoding for Motor Imagery BCI

August 2024 – December 2024

- Implemented EEGNet for 3-class hand grasp motor imagery classification (BCI Competition IV 2020 Track 4; 64.97% accuracy); benchmarked against 8 models spanning deep CNNs, and transformers under session-to-session EEG non-stationarity (50 trials/class, 7-day interval).

EDUCATION

University of Minnesota – Twin Cities

August 2024

Master of Science in Robotics (Specialization: Medical Robotics)

GPA: 3.95 / 4.00

Birla Institute of Technology and Science (BITS), Pilani

August 2018 – May 2023

Bachelor of Engineering in Mechanical Engineering

GPA: 8.37 / 10.00

Master of Science in Physics